

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

General comments

This manuscript addresses an interesting and theoretically important question: whether the exhaustive interpretation associated with Hungarian preverbal focus is semantically encoded or pragmatically inferred, and whether it can be empirically shown that a syntactic construction can imply contrast without it also implying exhaustiveness. The topic is clearly relevant to current debates in the syntax–semantics–pragmatics interface, and I appreciate the authors' experimental approach to the issue. The combination of behavioural choice data, self-paced reading, and eye-tracking is potentially valuable, and the paper engages with a substantial body of literature on structural focus in Hungarian and other languages.

However, in its present form, I do not think the study provides clear evidence for the theoretical conclusions the authors wish to draw. My main concern is methodological: the operationalisation of the contrastive/non-exhaustive interpretation using the phrase *többek között* 'among others' is difficult to separate from a much simpler inclusive interpretation in the sentence–picture matching task. Participants are asked to choose between a picture containing a single object and a picture containing the target object together with other objects. In this design, responses may be driven primarily by the referential expression itself — *only coffee*, *coffee*, or *coffee among other things* — rather than by the interpretation of Hungarian preverbal focus.

The problem arises partly because the pictures do not depict the result of the event described by the sentence. For example, they show coffee alone or coffee together with other drinks, rather than an event or state of affairs corresponding to "Eszter drank coffee" or "Eszter drank coffee among other things". As a result, the verb and the syntactic position of the object may be largely irrelevant to the task. The study therefore does not yet show that the contrast between exhaustive and non-exhaustive interpretations is due to the preverbal focus construction itself. A useful control would be a version of the task in which the same referential expressions are presented without the sentential structure, for example *only coffee*, *coffee*, and *coffee among other things*. If the same pattern of picture choices, decision times, and gaze behaviour were obtained in such a control, this would suggest that the present results are not specifically about Hungarian preverbal focus.

Method

There are also important concerns about the linguistic stimulus design and experimental procedure:

- The exhaustive and unmodified conditions begin with expressions such as "regarding drinks/fruits", which themselves evoke a set of alternatives and may encourage a contrastive or exhaustive interpretation. More importantly, the contrastive condition appears to differ from the other two conditions in broader sentence structure, discourse framing and lexical content. This creates a possible confound: differences between conditions may reflect differences in sentence context, lexical material or length, not only differences in focus interpretation.
- Other properties of the linguistic materials should be described and justified more fully such as the significance of the presence or absence of verbal prefixes, definite versus indefinite conjugation, and countable versus uncountable nouns so that these decisions are comprehensible to non-Hungarian speakers. Since the introduction presents preverbal focus

partly through the separation of the verbal prefix from the verb, it is also important to explain how stimuli without verbal prefixes fit into the design.

- The description of the experimental procedure also needs substantial clarification. It is unclear what exactly is meant by a “submission”, whether some participants completed more than one part of the experiment, whether they did so in a single sitting or separate sittings, whether this counted as one submission or two or three submissions.
- If the same participant could contribute more than one submission, this should be explicitly modelled. This is especially important if some participants may have seen the same items in different conditions since repeated exposure tends to have a major effect on reading and reaction times.
- The attention-check items should be described.
- Page 9, line 8: The phrase “the image types were counterbalanced throughout the trials” should be clarified; I assume this refers to the left/right position of the single-object and multi-object images, but this should be stated explicitly.
- Participant age and gender are reported, but the method section should state when and how these data were collected.

Data preprocessing

The preprocessing decisions for the eye-tracking data also need revision or stronger justification.

- On page 11, the authors state that long episodes were winsorised by replacing values above the mean + 2 SD with the global mean. This is unusual and may distort the data, because it replaces extreme values not with the nearest non-extreme boundary but with a central value. A more standard winsorisation procedure would replace values above the threshold with the threshold value itself.
- The text also uses inconsistent notation, referring to mean + 2 SD but also showing a ± 2 formula. It should be made clear whether only high outliers or both high and low outliers were treated.
- The notation for standard deviation is also inconsistent (SD vs sigma), the authors should state whether sample (SD) or population (sigma) standard deviation was used.
- If the position of image types was varied, it would be more meaningful to define AOIs with reference to image type rather than screen half.

Statistical analyses and results

The statistical analysis section should be expanded. The manuscript defines the models used for the eye-tracking measures, but the models for picture choice, sentence reading times, region-level reading times, and decision reaction times should also be stated explicitly. The authors should specify the contrast coding used for all categorical predictors. This is particularly important because the interpretation of fixed-effect coefficients and interactions depends on whether treatment coding or sum coding was used.

For the choice data, only descriptive statistics are reported. Given that picture choice is one of the central dependent variables, a statistical analysis should be added. Since the outcome is binary, a generalized linear mixed-effects model with a binomial family and logit link would be appropriate, with sentence condition as a fixed effect and random effects for participants and items.

The reading-time analyses require more caution. Whole-sentence reading times are difficult to interpret because the sentences are not matched for length. The exhaustive condition contains

csak, a one-syllable word, whereas the corresponding material in the unmodified condition varies in length, and the contrastive condition differs in several additional ways. The finding that exhaustive sentences are read faster may therefore simply reflect sentence length or lexical repetition, rather than easier semantic/pragmatic processing. If whole-sentence reading times are retained, the authors should control for sentence length, for example by including syllable count or character count as a covariate. Even then, lexical repetition remains a concern.

The same issue applies to Region 5. In the exhaustive condition this region contains *csak*, in the contrastive condition it contains part of the repeated phrase *többek között*, and in the unmodified condition it contains variable lexical material. Differences at this region are therefore confounded with word length, lexical identity, and repetition. I do not think strong theoretical conclusions about exhaustivity can be drawn from this region as currently analysed.

Regions 6 and 7 may be more informative because the critical object and final verb are more directly comparable across conditions. However, even here, possible spillover from the preceding region should be acknowledged. There is also a specific interpretive problem in the discussion of Region 6. The authors state that the effect of choice reflects slower reading when participants went on to select the multi-object image. However, Table 11 reports a negative coefficient for Choice B, with Choice A as the reference level. If the dependent variable is log reading time, this coefficient indicates shorter, not longer, reading times for Choice B in the reference condition. This should be corrected. More generally, because picture choice is highly imbalanced across sentence conditions — with multi-object choices rare in the exhaustive and unmodified conditions but frequent in the contrastive condition — choice effects are very unreliable. Picture choice is not an experimentally manipulated factor but an outcome of interpretation, and it is therefore not independent of sentence condition.

For all of the above analyses, as well as the decision time models, the inclusion of pairwise comparisons in general also needs careful consideration. When an interaction - as shown by the Chi square tests comparing models - does not even approach the level of significance, it is not appropriate to base substantive conclusions on separate pairwise comparisons within each level of the other factor, especially when the cell sizes are extremely imbalanced. In the exhaustive and unmodified conditions, multi-object choices are very rare; comparisons involving those trials have much larger standard errors and much lower power. A non-significant contrast in these cells cannot be interpreted as evidence that an effect is absent. The relevant test is the Sentence Condition × Choice Type interaction. Similarly, it is misleading to conclude that an effect is “limited to A-choice trials” merely because it is significant in the A subset and non-significant in the B subset. Such a conclusion requires a significant interaction.

I also have concerns about the eye-tracking measures. The webcam-based method is understandably less precise than laboratory eye-tracking, and the authors acknowledge this limitation. However, the analyses focus on total dwell time and total fixation count across the areas of interest. These measures are likely to correlate very strongly with decision time, because a participant who takes longer to decide will generally also spend longer looking at the images and may make more fixations. It is therefore not clear what independent theoretical contribution these measures make beyond the reaction-time data. If the authors wish to argue that the eye-tracking data show the simultaneous availability of two interpretations, they need measures that are more directly tied to competition between interpretations, such as looks to the single-object versus multi-object image over time, first fixation, switching patterns, or target-preference curves. Total

dwell time on the images as a whole does not by itself show that two interpretations are equally plausible.

Discussion

For the reasons detailed above, unfortunately no clear conclusions can be drawn from the data regarding the interpretation of preverbal focus constructions. I would encourage the authors to revise the manuscript substantially. At minimum, this would require: adding a statistical analysis of picture choice; clearly specifying all models and contrast coding; reanalysing reading times with appropriate controls for length and lexical differences; correcting the interpretation of the Region 6 choice effect; avoiding conclusions based on unequal pairwise comparisons where the interaction is not significant; and explaining what the eye-tracking measures contribute beyond decision time. Ideally, the authors should also include a control experiment or baseline condition without sentential structure, to test whether the same picture-choice pattern arises from the expressions *only X*, *X*, and *X among others* alone. Without such a control, it remains unclear whether the study is testing preverbal focus interpretation or simply the interpretation of the overt modifiers used in the task.

Terminology and typos

Finally, I suggest some terminological and presentation revisions.

- The task is better described as a sentence–picture matching or picture-selection task rather than a sentence–picture verification task, since participants choose between two pictures rather than verifying whether a single picture matches the sentence.
- The image alternatives should be named consistently throughout rather than alternating between single-object/multi-object, exhaustive/contrastive, and type A/type B.
- Abstract: “central decade” should be “central debate”.
- Abstract: “exhaustitive” should be “exhaustive”.
- Page 5: “due the use” should be “due to the use”.
- Page 9: “advance throught the sentences” should be “advance through the sentences”.
- Page 9: “in one setting” should be “in one sitting”.
- Page 9: “where there faces” should be “where their faces”.
- Page 10: “bad lighing” should be “bad lighting”.
- Page 10: “intstructed” should be “instructed”.
- Page 10: “they could began” should be “they could begin”.
- Tables and figure captions should be checked for consistency, especially the terminology “fixation times” versus “fixation counts”.